Math 161
Problem set 6 solutions

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1. Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) := \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ a & \text{if } x \notin \mathbb{Q}. \end{cases}$$

To see that f is continuous at a, first observe that $a \leq f(x) \leq x$ for $x > a$. Since

$$\lim_{x \to a^+} a = a = \lim_{x \to a^+} x,$$

the squeeze theorem (exercise 5 on Problem Set 4) implies that $\lim_{x \to a^+} f(x) = a$ (the exercise was stated for two-sided limits, but the same proof shows that it holds for one-sided limits as well; alternatively, rewrite the one-sided limits as two-sided limits using absolute values). Similarly, we have $x \leq f(x) \leq a$ for $x < a$, and since

$$\lim_{x \to a^-} x = a = \lim_{x \to a^-} a,$$

it follows that $\lim_{x \to a^-} f(x) = a$. Applying exercise 1 from the previous problem set, we obtain

$$\lim_{x \to a} f(x) = a = f(a)$$

as needed.

Now we must prove that f is discontinuous at b for any $b \neq a$. For this, we have to show that there exists $\epsilon > 0$ such that for any $\delta > 0$, there exists $x \in \mathbb{R}$ such that $|x - b| < \delta$ and $|f(x) - f(b)| \geq \epsilon$. Let $\epsilon := \frac{1}{2}|a - b|$ and fix $\delta > 0$. Put $\delta' := \min(\epsilon, \delta)$ and choose $x \in \mathbb{R}$ such that $|x - b| < \delta'$ and

- if $b \in \mathbb{Q}$, then $x \notin \mathbb{Q}$;
- if $b \notin \mathbb{Q}$, then $x \in \mathbb{Q}$.

It is possible to choose such an x because both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$. Note that $|x - b| < \delta' \leq \delta$, so one of the desired inequalities is satisfied. For the other, we separate into cases: if $b \in \mathbb{Q}$, then $x \notin \mathbb{Q}$ and hence $f(b) = b$, $f(x) = a$. Thus

$$|f(x) - f(b)| = |a - b| > \tfrac{1}{2}|a - b| = \epsilon$$

as needed. If $b \notin \mathbb{Q}$, then $x \in \mathbb{Q}$ and hence $f(b) = a$, $f(x) = x$. It follows that

$$\begin{aligned} |f(x) - f(b)| &= |x - a| = |a - x| \\ &= |a - b - (x - b)| \geq |a - b| - |x - b| \\ &> |a - b| - \delta' \geq |a - b| - \epsilon \\ &= \epsilon. \end{aligned}$$

2. Let $g : [0, 1] \to \mathbb{R}$ be the function defined by $g(x) := f(x) - x$. Since f and the function $x \mapsto x$ are continuous, so is their difference g. Note that $f(0) \geq 0$ implies that $g(0) = f(0) - 0 \geq 0$. Similarly, the inequality $f(1) \leq 1$ implies that $g(1) = f(1) - 1 \leq 0$. Thus, by the intermediate value theorem, there exists $x \in [0, 1]$ such that $g(x) = 0$, i.e. $f(x) = x$.

3. Assuming that $A \subset \mathbb{Q}$, we prove the contrapositive: if A does not have exactly one element, then A is disconnected. If $A = \varnothing$, then certainly A is disconnected. If A has more than one element, then there exist $x, y \in A$ such that $x < y$. Since $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$, there exists $z \in \mathbb{R} \setminus \mathbb{Q}$ such that $x < z < y$. Let $U := (-\infty, z)$ and $V := (z, \infty)$. Now $z \notin \mathbb{Q}$ implies that $z \notin A$ and hence

$$A \subset \mathbb{R} \setminus \{z\} = U \cup V.$$

Since $U \cap V$, in particular we have $A \cap U \cap V = \varnothing$. Finally, we have $x \in A \cap U$ and $y \in A \cap V$, so $A \cap U \neq \varnothing$ and $A \cap V \neq \varnothing$. Thus A is disconnected as claimed.

4. (a) The definitions are as follows.
   - We write $\lim_{x\to\infty} g(x) = \infty$ if for any $M > 0$, there exists $N > 0$ such that $x > N$ implies that $g(x) > M$ for $x \in \mathbb{R}$.
   - We write $\lim_{x\to\infty} g(x) = -\infty$ if for any $M > 0$, there exists $N > 0$ such that $x > N$ implies that $g(x) < -M$.
   - We write $\lim_{x\to-\infty} g(x) = \infty$ if for any $M > 0$, there exists $N > 0$ such that $x < -N$ implies that $g(x) > M$.
   - We write $\lim_{x\to-\infty} g(x) = -\infty$ if for any $M > 0$, there exists $N > 0$ such that $x < -N$ implies that $g(x) < -M$.

   It is not important here that we use strict inequalities.

   (b) To prove that $\lim_{x\to\infty} x^n = \infty$, we must show that for any $M > 0$, there exists $N > 0$ such that $x > N$ implies that $x^n > M$. Given $M > 0$, let $N := \max(M, 1)$. If $x > N$, then it follows that $x^n > N^n$ (since $N > 0$). We have $N \geq 1$, and hence $N^n \geq N$. Finally, we have $N \geq M$, so we conclude that $x^n > M$ as desired.

   As for $\lim_{x\to\infty} x^n = -\infty$, given $M > 0$, let $N := \max(M, 1)$ as above. We have seen that then $x > N$ implies that $x^n > M$. If $x < -N$, then $-x > N$, and since n is odd we have

$$-x^n = (-x)^n > M.$$

   Thus $x^n < -M$ as needed.

   (c) We'll use the following lemma: if $\lim_{x\to\infty} g(x) = \infty$ and $\lim_{x\to\infty} h(x) = L > 0$, then

$$\lim_{x\to\infty} g(x)h(x) = \infty.$$

   Fix $M > 0$, so we must show that there exists $N > 0$ such that $x > N$ implies that $g(x)h(x) > M$. Since $\lim_{x\to\infty} h(x) = L > 0$, there exists $N_1 > 0$ such that $x > N_1$ implies that $|h(x) - L| < \frac{L}{2}$, and hence

$$h(x) > L - \frac{L}{2} = \frac{L}{2}.$$

   On the other hand, since $\lim_{x\to\infty} g(x) = \infty$, there exists $N_2 > 0$ such that $x > N_2$ implies that $g(x) > \frac{2M}{L}$. Put $N := \max(N_1, N_2)$, so that $x > N \geq N_1, N_2$ implies that

$$g(x)h(x) > \frac{2M}{L} \cdot \frac{L}{2} = M.$$

   Likewise, one proves similarly that if $\lim_{x\to\infty} g(x) = -\infty$ and $\lim_{x\to\infty} h(x) = L > 0$, then

$$\lim_{x\to\infty} g(x)h(x) = -\infty.$$

   Next, we observe that

$$\lim_{x\to\infty} \left(1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_1}{x^{n-1}} + \frac{a_0}{x^n}\right) = \lim_{x\to 0^+} (1 + a_{n-1}x + \cdots + a_1 x^{n-1} + a_0 x^n) = 1,$$

where the first equality uses Exercise 3(a) on Problem Set 5, and the second equality is because polynomials are continuous. Since the factorization

$$f(x) = x^n \left(1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_1}{x^{n-1}} + \frac{a_0}{x^n}\right)$$

holds for $x \neq 0$ and in particular for x sufficiently positive, the lemma implies that $\lim_{x\to\infty} f(x) = \infty$ and $\lim_{x\to-\infty} f(x) = -\infty$.

(d) Since $\lim_{x\to-\infty} f(x) = -\infty$, there exists $N_1 > 0$ such that $x < -N_1$ implies that $f(x) < -1$. In particular, we have e.g.

$$f(-N_1 - 1) < -1 < 0.$$

Similarly, since $\lim_{x\to-\infty} f(x) = -\infty$, there exists $N_2 > 0$ such that $x > N_2$ implies that $f(x) > 1$, and in particular

$$f(N_2 + 1) > 1 > 0.$$

Since f is a polynomial and therefore continuous, the intermediate value theorem implies that there exists $x \in [-N_1 - 1, N_2 + 1]$ such that $f(x) = 0$.

5. If $A = \varnothing$ then we are already done, since $\varnothing$ appears on the list. So we assume below that $A \neq \varnothing$.

The set A is either bounded above, bounded below, both, or neither. Suppose first that A is unbounded above and below. We claim that in that case $A = \mathbb{R}$. Namely, if $x \in \mathbb{R}$ then there exist $y, z \in A$ such that $y < x < z$, since A is unbounded above and below. Since A is an interval, it follows that $x \in A$. Thus we have shown that $\mathbb{R} \subset A$, and since $A \subset \mathbb{R}$ by hypothesis, we have $A = \mathbb{R}$.

Next, suppose that A is bounded above but not below. Since $A \neq \varnothing$, the least upper bound property implies that $b := \sup A$ exists. We claim that if $b \in A$, then $A = (-\infty, b]$. Since b is an upper bound for A, we have $A \subset (-\infty, b]$. On the other hand, if $x \in (-\infty, b]$, then there exists $y \in A$ such that $y < x$ because A is unbounded below. Thus $y < x \leq b$, and since $b, y \in A$ and A is an interval, we deduce that $x \in A$. This shows that $(\infty, b] \subset A$ and hence $A = (-\infty]$ as claimed.

Continuing to assume that A is bounded above but not below, suppose that $b = \sup A \notin A$. Then we claim that $A = (\infty, b)$. If $x \in A$, then $x < b$ because b is an upper bound for A and $b \notin A$, which shows that $A \subset (-\infty, b)$. For the other inclusion $(-\infty, b) \subset A$, suppose that $x \in (-\infty, b)$. Since A is unbounded below, there exists $y \in A$ such that $y < x$. On the other hand, there exists $z \in A$ such that $x < z$, since otherwise x would be an upper bound for A, which is impossible because $x < b$. Now we have $y < x < z$, which implies that $x \in A$ because $y, z \in A$ and A is an interval. We conclude that $(-\infty, b) \subset A$ and hence $A = (-\infty, b)$.

Next, we consider the case where A is bounded below but not above. In particular $a := \inf A$ exists. One can repeat the arguments from the previous two paragraphs with inequalities reversed to show that $A = [a, \infty)$ or $A = (a, \infty)$. Alternatively, apply exercise 1(b) on Problem Set 3 to see that $-a = \sup(-A)$, and since $-A$ is clearly an interval, the above arguments show that $-A = (-\infty, -a]$ or $-A = (-\infty, -a)$. Thus $A = [a, \infty)$ or $A = (a, \infty)$.

Now assume that A is bounded (i.e. bounded above and below). It follows that $a := \inf A$ and $b := \sup A$ both exist. If $a, b \in A$, then we claim that $A = [a, b]$. Since a and b are lower and upper bounds for A respectively, we have $A \subset [a, b]$. To see the other inclusion, suppose that $a \leq x \leq b$. Since A is an interval, it follows that $x \in A$, which shows that $[a, b] \subset A$ and hence $A = [a, b]$.

Next, suppose that A is bounded but $a, b \notin A$. We claim that $A = (a, b)$. Since a and b are lower and upper bounds for A respectively, for any $x \in A$ we have $a \leq x \leq b$. But $x \neq a, b$ because $a, b \notin A$, so $a < x < b$ and hence $A \subset (a, b)$. Conversely, suppose that $a < x < b$. It follows that there exists $y \in A$ such that $y < x$, since otherwise x would be a lower bound for A, which is impossible because $a < x$. Similarly, there exists $z \in A$ such that $x < z$, because otherwise x would be a lower bound for A in contradiction of $x < b$. Now $y < x < z$ implies that $x \in A$ because $y, z \in A$ and A is an interval. This shows that $(a, b) \subset A$ and hence $A = (a, b)$.

Continuing to assume that A is bounded, suppose that $a \in A$ but $b \notin A$. Then we claim that $A = [a, b)$. Since a and b are lower and upper bounds for A respectively and $b \notin A$, we have $A \subset [a, b)$. If $a \leq x < b$, then there exists $y \in A$ such that $x < y$ because $b = \sup A$ and $x < b$. Now $a \leq x < y$ implies that $x \in A$ because $a, y \in A$ and A is an interval. Thus $[a, b) \subset A$ and hence $A = [a, b)$.

Finally, suppose that A is bounded and $a \notin A$, $b \in A$. One can repeat the previous paragraph with inequalities reversed to show that $A = (a, b]$. Alternatively, note that $-A$ is an interval and $-a = \sup(-A)$, $-b = \inf(-B)$ by exercise 1(b) from Problem Set 3. Since $-a \notin A$, $-b \in B$, the previous paragraph says that $-A = [-b, -a)$ and hence $A = (a, b]$ as desired.